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TRANSFLOW CHECKOUT PAYMENT

API DEFINITION

CONFIDENTIAL

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INTENDED AUDIENCE

This document is intended for the following audience:

1. System integrators
2. Application developers
3. Project Officers

SCOPE

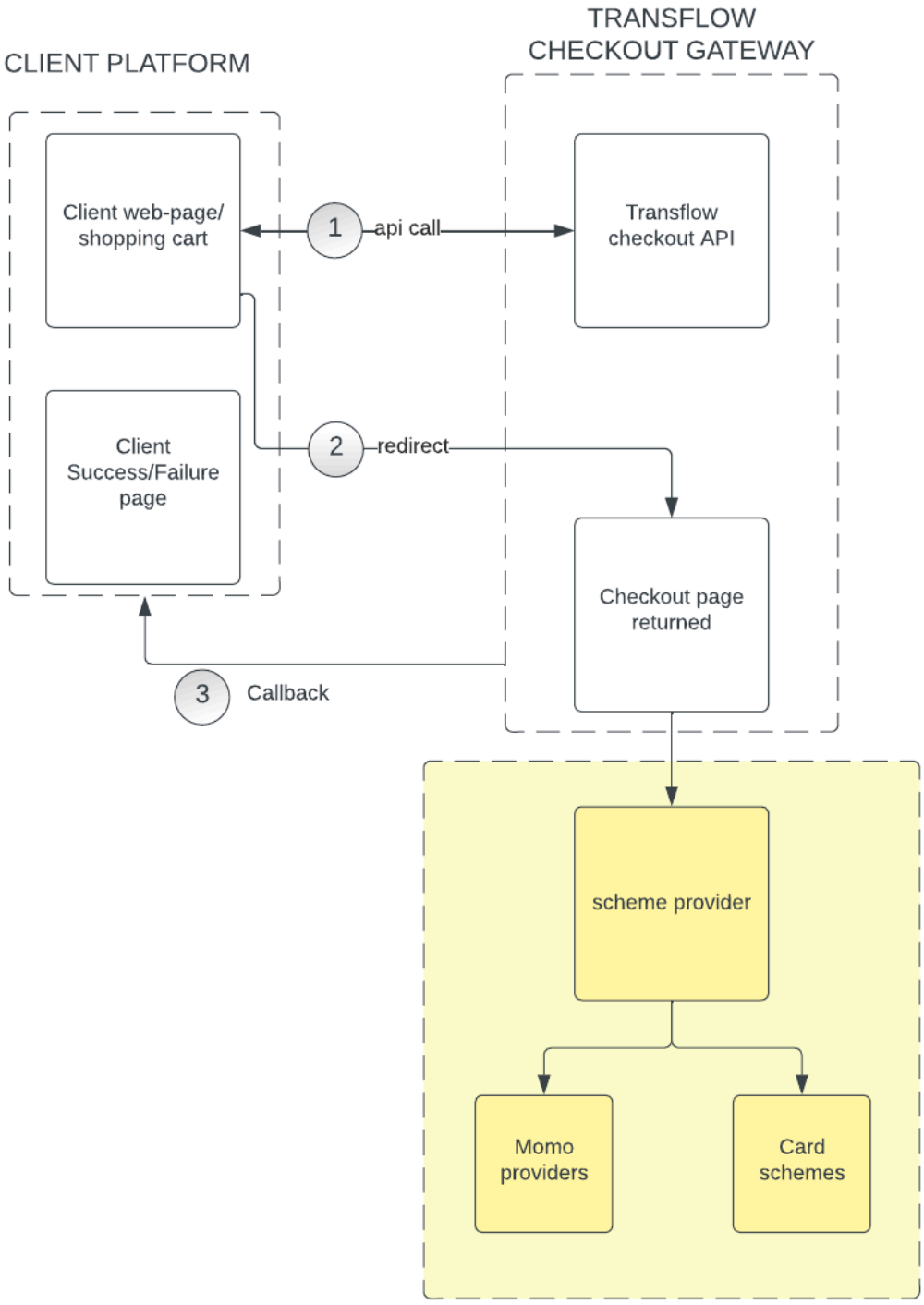
This document outlines a set of API methods designed for merchants to seamlessly process both mobile money (across all telco networks in Ghana) and card payments via a web checkout page. It includes details on the API name, input and output parameters, and intended use, serving as a guide for System Integration Tests (SIT). Application developers can utilize this document to integrate with our payment gateway service.

Base URL

https://apisuat.itcsrvc.com/checkout - UAT

https://apis.itcsrvc.com/checkout - LIVE

PAYMENT FLOW



1. The client on gathering data from shopping cart/payment page, will initiate a call to the transflow checkout API
2. The client API will redirect to the checkout page and require the client to authorize payment
3. Upon successful completion or otherwise, the checkout gateway will send the callback to a designated success/failure page on the client side

1. PAYMENT REQUEST INITIATION

This Transflow checkout API allows you to initiate payments. You are required to pass information about the payment you are about to make so that on the checkout page, that information is pre-populated for you. The response from the API is a full url that the implementor will redirect to, for the completion of payment.

API METHOD NAME	/request-payments
INTENT	To receive payment detail of the requester
Method	POST
Content Type	application/json

Request			
Parameter	**Type**	**Mandatory**	**Comment**
fullName	String	Y	The name of the customer initiating payment.
email	String	Y	The customer's email address is a required field when making a card payment.
paymentMethod	String	N (“card”,“mobile_money”)	The payment method field is optional. If a customer is making a payment using a card, they should select **'card'** as the option for the payment method field. However, if a customer

			is making a mobile money transaction, they should select **'mobile_money'** as the value for the payment method field
narration	String	Y	Payment narration
amount	Number	Y	The amount transacted
currency	String	Y	The currency in which transaction is being done
logo	String	N	The url link to the merchant/institution logo
successRedirectUrl	String	Y	This is the URL to which customers will be redirected upon successful completion of a transaction
failureRedirectUrl	String	Y	This is the URL to which customers will be redirected when a transaction is unsuccessful
pageTitle	String	Y	Custom page title the merchant would like to display on the checkout page
pageDescription	String	N	A custom page description that the merchant wishes to display on the checkout page.
msisdn	String	N	Phone number is required if payment method is **mobile_money**
apiKey	String	Y	This is the authentication parameter required in the request.
merchantProductId	String	Y	Identifier for the product

transflowId	String	Y	Client transflowId which is required in the request.

Response			
Parameter	**Type**	**Mandatory**	**Comment**
responseCode	Number	Y	A numeric code to represent the response
responseMessage	String	Y	Explanation of the numeric code
transactionReference	String	Y	This is the reference for the payment request. Implementers should store this reference. Callbacks sent after payment will come with this reference String
paymentLink	String	Y	This is the URL to the checkout-page with the relevant details. Clients can be redirected to the page for payments

SAMPLE REQUEST (REQUEST PAYMENT)

Headers

```
Unset
Content-Type: application/json
```

Body

```
Unset
{
    "fullName": "william mensah",
    "email": "williammensah@gmail.com",
    "paymentMethod": "card",
    "narration": "This is a generic payment description",
     "amount": 200,
     "currency":"GHS",
     "transflowId": "xxxxxxxxx",
     "merchantProductId": "xxxxxxxxx",
     "apiKey": "xxxxxxxxxxx"
     "logo": "https://www.knust.edu.gh",
     "successRedirectUrl": "https://www.knust.edu.gh/",
     "failureRedirectUrl": "https://www.knust.edu.gh/",
     "callbackUrl": "https://www.knust.edu.gh",
     "pageDescription": "Knust Checkout Page",
     "pageTitle": "Knust",
    "currency": "GHS",
    "msisdn": "233201991407",
    "network": "VODAFONE"
}
```

Response `200`

Headers

```
Unset
Content-Type: application/json
```

Body

Unset


```
{
    "responseCode": 200,
    "responseMessage": "Success",
    "data": {
        "transactionReference":
"67ac03ec-3132-4e9c-93e4-1a78bb942355",
        "checkoutUrl":
"https://checkoutuat.transflowitc.com?req=eyJmdWxsTmFtZSI6IndpbGx
pYW0gbWVuc2FoIiwiZW1haWwiOiJ3aWxsaWFtbWVuc2FoQGdtYWlsLmNvbSIsInBh
eW1lbnRNZXRob2QiOiJjYXJkIiwibmFycmF0aW9uIjoiVGhpcyBpcyBhIGdlbmVya
WMgcGF5bWVudCBkZXNjcmlwdGlvbiIsImFtb3VudCI6MjAwLCJsb2dvIjoiaHR0cH
M6Ly93d3cua251c3QuZWR1LmdoIiwicGFnZURlc2NyaXB0aW9uIjoiS251c3QgQ2h
lY2tvdXQgUGFnZSIsInBhZ2VUaXRsZSI6IktudXN0IiwiY3VycmVuY3kiOiJHSFMi
LCJtc2lzZG4iOiIyMzMyMDE5OTE0MDciLCJuZXR3b3JrIjoiVk9EQUZPTkUiLCJ0c
mFuc2FjdGlvblJlZmVyZW5jZSI6IjY3YWMwM2VjLTMxMzItNGU5Yy05M2U0LTFhNz
hiYjk0MjM1NSJ9"
    }
}
```


2. PAYMENT CALLBACKS

After payments are made on the platform, the user will be redirected to either the success or failure url that was submitted in the initial request for payments.
The requestor will also receive a callback from the service backend to let the user know of the state of payments. The implementer of the api should respond to every request with Http Status Code 200, and a json response as below.

JavaScript

Implementers are advised to store the callback response
and return the required response even in the event of failure so that they can internally handle the processing of their transactions and not rely on the external service if the callback has been delivered.

The sample callback that is sent for successful and failed transactions can be found below;

Success:

```JavaScript
{

uniwalletTransactionId:'739d77c9-f73e-4dfd-b05a-0372790ab242',
    networkTransactionId: '39d8f4d0-6bfc-461a-ad25-97eb075bea5b',
    refNo: 'c799e1bd-de3f-4520-8871-7d63373a13f6',
    productId: '87220cfe-b77e-454a-aba7-f06a4056f14f',
    msisdn: '233201991407',
    amount: '12.00',
    narration: 'william',
    timestamp: '2023-07-02 18:10:53',
    responseCode: '01',
    responseMessage: 'Transaction processed successfully',
    network: 'VODAFONE',
    transflowId: '0da769f5-202b-485c-8818-306b9364f76d'
  }
```

Failure:

```javascript
{
  uniwalletTransactionId: 'c665f850-5cd7-4b7f-b7df-cb5eaf1f163a',
  networkTransactionId: '',
  refNo: '890f7f8b-f1f3-4d82-8909-e8d6ed3a2a6e',
  productId: 'e50cc86c-fde1-4eb7-882c-5c5f05b06642',
  msisdn: '233201991407',
  amount: '12.00',
  narration: 'This is  mobile payment description',
  timestamp: '2023-07-02 18:07:58',
  responseCode: '100',
  responseMessage: 'Transaction could not be completed at this
time. Please try again later',
  network: 'VODAFONE',
  transflowId: '0da769f5-202b-485c-8818-306b9364f76d'
}
```

3. CHECK TRANSACTION STATUS

There will be occasions where implementers do not receive callback after a stipulated period and would like to check the status of the transaction in order to deliver service to their customer.

API METHOD NAME	/check-transaction-status
INTENT	Check the status of a transaction
Method	POST
Content Type	application/json

Request			
Parameter	**Type**	**Mandatory**	**Comment**
transactionReference	String	Y	This is the reference for the payment request.
apiKey	String	Y	This is the authentication parameter required in the request.
merchantProductId	String	Y	Identifier for the product
transflowId	String	Y	Client transflowId which is required in the request.

Response			
Parameter	**Type**	**Mandatory**	**Comment**
responseCode	Number	Y	A numeric code to represent the response
responseMessage	String	Y	Explanation of the numeric code
data	Object	Y	This contains details of the payment

SAMPLE REQUEST (CHECK TRANSACTION STATUS)

Headers:

```
Unset
Content-Type: application/json
```

Body:

```
Unset
{
    "apiKey":"xxxxxxxxxxxxxxxxxxxxxxxxxxxx",
    "merchantProductId": "xxxxxxxxxxxxxxxxxxxx",
    "transflowId": "xxxxxxxxxxxxxxxxxxxx",
    "transactionReference":"0003ca4d-8c7a-4f42-90ec-bd7b0a54ff67"
}
```

Response `200`

Headers:

```
Unset
Content-Type: application/json
```

Body:

```
JavaScript
{
    "responseCode": 200,
    "responseMessage": "Success",
    "data": {
        "timestamp": "2023-06-21T04:33:33.000Z",
        "currency": "GHS",
        "amount": 17,
        "refNo": "0003ca4d-8c7a-4f42-90ec-bd7b0a54ff67",
        "uniwalletTransactionId":
"ccc42dad-c4f0-401c-8d1e-d5c382304676",
        "responseCode": "01",
        "responseMessage": "Operation Successful"
    }
}
```

4. GET PAYMENT BREAKDOWN

This endpoint is designed to fetch the potential fee charges applicable to the upcoming transaction initiated by the client.

API METHOD NAME	/payment-breakdown
INTENT	To receive retrieve payment fees
Method	POST
Content Type	application/json

Request			
Parameter	**Type**	**Mandatory**	**Comment**
transactionReference	String	Y	This is the reference for the payment request.
apiKey	String	Y	This is the authentication parameter required in the request.
merchantProductId	String	Y	Identifier for the product
transflowId	String	Y	Client transflowId which is required in the request.
currency	String	Y	Default GHS
amount	Number	Y	The amount transacted
paymentMethod	String	Y (“card”,“mobile_money”)	The mode of payments whether is mobile money or card transfer

network	String	Y	For card payments, pass network as “CARD”
accountReference	String	Y	This indicates the client's phone number. If the payment method is mobile money, please provide the phone number as the value. If the payment is made via card, use the card number as the account reference.

Response			
Parameter	**Type**	**Mandatory**	**Comment**
responseCode	Number	Y	A numeric code to represent the response
responseMessage	String	Y	Explanation of the numeric code
data	Object	Y	This contains details of the payment

SAMPLE REQUEST (PAYMENT BREAKDOWN)

Headers:

```
Unset
Content-Type: application/json
```

Body:

```
Unset
{
     "merchantProductId": "xxxxxxxxxxxxxxxxxxxxx",
     "currency": "GHS",
     "amount": 200,
     "paymentMethod": "mobile_money"
     "network":"MTN",
     "accountReference": "0246827533"

}
```

Response `200`

Headers:

```
Unset
Content-Type: application/json
```

Body:

```
JavaScript
{
    "responseCode": 200,
    "responseMessage": "Success",
    "data":{
        "channelFee": 30,
        "transactionAmount": 2000,
        "chargeableAmount": 2019.13,
        "chargeBearer": "merchant",
        "chargeMethod": "percentage",
        "transactionId": "a6e9a565-65f2-42e3-b6e5-a40ee0d861e5",
        "taxableAmount": 1913,
        "elevyTaxAmount": 19.13,
        "vatAmount": 0,
        "totalTaxAmount": 19.13
```

```
}
```

5. SEND OTP

This endpoint is used to send OTP for mobile money transactions.

API METHOD NAME	/send-otp
INTENT	To send OTPs.
Method	POST
Content Type	application/json

Request			
Parameter	**Type**	**Mandatory**	**Comment**
msisdn	String	Y	The client phone number

Response			
Parameter	**Type**	**Mandatory**	**Comment**
responseCode	Number	Y	A numeric code to represent the response
responseMessage	String	Y	Explanation of the numeric code

data	Object	Y	This contains details of the payment

SAMPLE REQUEST (SEND OTP)

Headers:

```
Unset
Content-Type: application/json
```

Body:

```
Unset
{
    "msisdn": "0550522027"
}
```

Response `200`

Headers:

```
Unset
Content-Type: application/json
```

Body:

```
JavaScript
{
    "responseCode": 201,
    "responseMessage": "Success",
    "data": {
        "Otp": "PDWD"
```

```
        }
    }
```

6. VERIFY OTP

This endpoint is used to verify OTP sent.

API METHOD NAME	/verify-otp
INTENT	To verify opt sent
Method	POST
Content Type	application/json

Request			
Parameter	**Type**	**Mandatory**	**Comment**
otp	String	Y	The OTP client entered
otp-code	String	Y	The OTP code

Response			
Parameter	**Type**	**Mandatory**	**Comment**
responseCode	Number	Y	A numeric code to represent the response
responseMessage	String	Y	Explanation of the

			numeric code
data	Object	Y	This contains details of the payment

SAMPLE REQUEST (VERIFY OTP)

Headers:

```
Unset
Content-Type: application/json
```

Body:

```
Unset
{
    "otp-code": "WFBS",
    "otp":"191093"
}
```

Response `200`

Headers:

```
Unset
Content-Type: application/json
```

Body:

```javascript
{
    "responseCode": 200,
    "responseMessage": "Otp verification has passed successfully",
    "data": {}
}
```

7. Make Payment (Socket Connection)

To start your payment, please connect to our secure server using this link:
wss://inzs8y2ti7.execute-api.eu-west-1.amazonaws.com/prod.
Once connected, be sure to listen for and respond to specific messages the server sends. It's important to act on these messages promptly.

However, if you don't receive any response within one minute, please close the connection and try again later. You will receive an email confirmation if the payment goes through, even if there's a temporary delay.

Here are some key points to remember:

- Use the provided link to establish a secure connection.
- Pay close attention to messages from the server and respond accordingly.
- If no response comes within a minute, disconnect and try again later.
- An email confirmation will be sent regardless of any temporary delays.

Socket Responses			
Parameter	**Type**	**Mandatory**	**Comment**
responseCode	Number	Y	For success payment responseCode is **201** and for failed payment

			responseCode is **500**
responseMessage	String	Y	Explanation of the numeric code
successRedirectUrl	String	Y	Success payment redirect url
failureRedirectUrl	String	Y	Failure payment redirect url

Request			
Parameter	**Type**	**Mandatory**	**Comment**
merchantProductId	String	Y	Identifier for the product
fullName	String	Y	The name of the customer initiating payment.
email	String	Y	The customer's email address is a required field when making payment.
route	String	Y	The default value is **external**. This helps indicate the route from which the payment is originating from
paymentMethod	String	Y	The payment method field.If a customer is making a payment using a card, they should select **'card'** as the option for the payment

			method field. However, if a customer is making a mobile money transaction, they should select **'mobile_money'** as the value for the payment method field
narration	String	Y	Payment narration
amount	String	Y	The initial amount provided by customer
fullAmount	String	Y	The amount plus e-levy charges
tokenId	String	N	To ensure card details are retained in Skyflow vaults during card payments, include the tokenId in the payment process.
countryOfIssue	String	N	Represents card country of issuer country code
pan	String	N	For card payments pan is required
cvv	String	N	For card payments cvv is required
cardNumber	String	N	For card payments, the cardNumber is required
currency	String	Y	The currency in which transaction is being done
msisdn	String	N	Phone number is required if payment method is **mobile_money**
network	String	Y	For card payments, pass network as “CARD”
expiryYear	String	N	For card payments the expiryYear is required in this format “39”
expiryMonth	String	N	For card payments the

			expiryMonth is required in this format “01”

8. Store Card Vaults

API METHOD NAME	/card-vaults
INTENT	To store card details in skyflow vaults
Method	POST
Content Type	application/json

This endpoints stores customer card details to card vaults for instances where customers want their card information to be stored

Card Vault Responses			
Parameter	**Type**	**Mandatory**	**Comment**
responseCode	Number	Y	For success payment responseCode is **200** and for failed payment responseCode is **500**
responseMessage	String	Y	Explanation of the numeric code

tokenId	String	Y	The unique identifier from Skyflow that represents the stored card information in Skyflow is the tokenId.
countryOfIssue	String	Y	Card number country of issue code
cardNumberSuffix	String	Y	The last four digits of the card number

Request			
Parameter	**Type**	**Mandatory**	**Comment**
merchantProductId	String	Y	Identifier for the product
cvv	String	Y	For card payments cvv is required
cardNumber	String	Y	For card payments, the cardNumber is required
apiKey	String	Y	This is the authentication parameter required in the request.
transflowId	String	Y	Client transflowId which is required in the request.
expiryYear	String	Y	For card payments the expiryYear is required in this format “39”
expiryMonth	String	Y	For card payments the expiryMonth is required in this format “01”

9. Remove Card From Vaults

The endpoints remove customer card details from card vaults.

API METHOD NAME	/remove-card
INTENT	To remove card details in skyflow vaults
Method	POST
Content Type	application/json

Request			
Parameter	**Type**	**Mandatory**	**Comment**
tokenId	String	Y	Identifier for skyflowId

Remove Card Responses			
Parameter	**Type**	**Mandatory**	**Comment**
responseCode	Number	Y	For success payment responseCode is **200** and for failed payment responseCode is **500**
responseMessage	String	Y	Explanation of the numeric code
skyflowId	String	Y	The unique identifier from Skyflow that represents the stored card information in Skyflow is the tokenId.

TEST DATA

For mobile transactions, you can use any registered number to test mobile payments. The amounts to be used for the testing should be small values eg GHS 0.1.

For card transactions, you can use the following cards to get the different responses.

SUCCESS:
PAN: 5123450000000008
CVV: 100
EXPIRY: 01/39

FAILURE:
PAN: 3528000000000007
CVV: 101
EXPIRY: 05/39